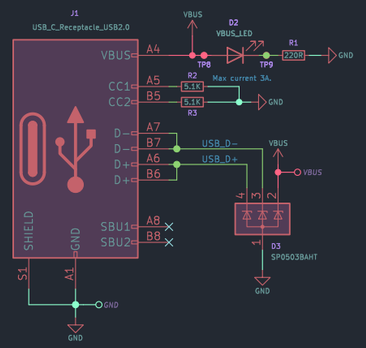
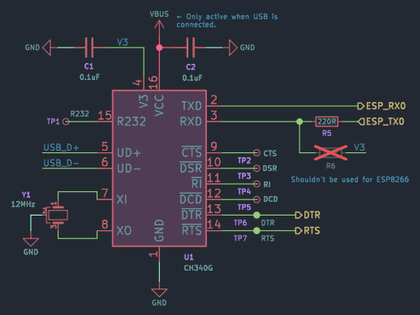
USB-C CONNECTOR

16-pin USB-C receptacle in USB 2.0 mode. Max current 3A.



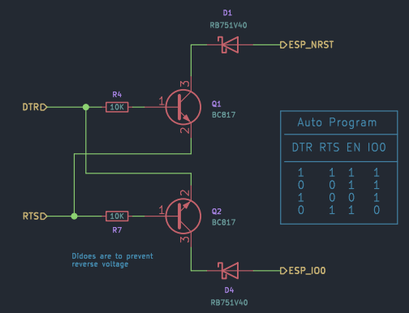
CH340G

USB 2.0 to serial converter.



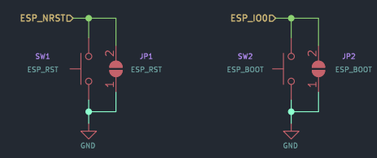
AUTO-PROGRAM

Auto reset and boot mode circuit. Uses DTR and RTS signals from the USB-Serial converter.



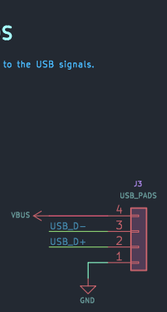
PUSH-BUTTONS

ESP reset and boot mode push-buttons for manual operation.
Also include solder points for external switches.



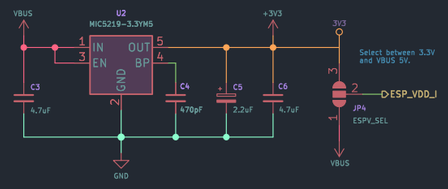
USB-C PADS

Allows soldering wires to the USB signals.



3.3V LDO

3.6-12V in, 3.3V out, 0.6V @ 500mA, PSRR 75dB (120Hz)

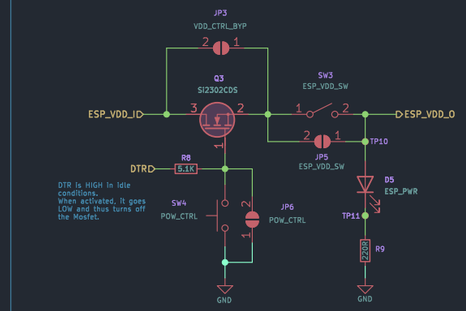


CONNECTOR - HEADER

Generic connector for connecting to the target ESP.

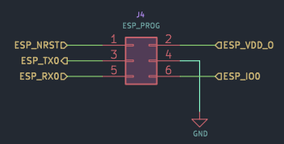
ESP VDD CONTROL

If you don't have access to the ESP NRST pin, you can use the DTR signal to disconnect the power to the ESP by removing the jumper and allowing the Mosfet to control the supply to the ESP.



CONNECTOR - ESP-PROG

Allows to use the official ESP-Prog cables.



MOUNTING HOLES

Mechanical mounting holes.

CIRCUITSTATE
WWW.CIRCUITSTATE.COM

https://www.circuitstate.com

GitHub: https://github.com/CIRCUITSTATE/ESPNut-D1

Licence: MIT Open-Source Licence

ESP32 & ESP8266 Serial Auto-Programmer
Designed by: Vishnu Mohanan (@vishnumaiea, @vizmohanan)
CIRCUITSTATE Electronics LLP (www.circuitstate.com)

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File: ESPNut-D1.kicad_sch

Title: ESPNut-D1

Size: A3	Date: 2025-04-23	**Rev: 0.1**
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